GGG–E–936c
February 24, 1967
SUPERSEDING
Int. Fed. Spec. GGG–E–00936b (GSA–FSS)
July 14, 1965 and
Fed. Spec. GGG–E–936a
January 31, 1964

FEDERAL SPECIFICATION

# EXTRACTOR, SCREW (STUD AND PIPE), AND EXTRACTOR, TAP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated extractors for use in removing broken screws, studs, bolts, pipes, and taps.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Extractors, covered by this specification, shall be of the following types, classes, and styles, as specified (see 6.2):

Type I—Screw and pipe extractors.

Class 1—Tapered, spiral flutes or ridges.

Styles A—Drill.
B—Plug.

Class 2—Tapered, straight fluted.

Class 3—Straight, straight fluted.

Type II—Tap extractors.

Type III—Screw extractors.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

GGG–D–751—Drills, Twist (Taper-Square, Straight, and Taper-Round Shanks).

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals; Test Methods.

Fed. Test Method Std. No. 406—Plastics: Methods of Testing.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–H–15424—Hand Tools; Packaging of.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of extractors which are otherwise in accordance with the requirements of the specification.

3.2 Material. Material used in the manufacture of extractors shall be steel, and shall be free from all defects and imperfections that may adversely affect the serviceability of the finished product.

3.3 Finish. The surface characteristics and finish shall be in conformance with high quality, commercial practice.

3.4 Identification marking. Each extractor or turnout nut shall be marked in a permanent and legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. In addition, the following shall be marked in a permanently legible manner on applicable items:

Recommended drill size on type I, class 1, styles A and B; type I, class 2; type I, class 3; type III, sizes 6 through 8 extractors; and on the turnout nuts of type III, sizes 1 through 5 extractors; and on the drill guides as shown in table VIII.

Nominal outside diameter on drill guides for type III, sizes 1 through 5 extractors.

Nominal pipe size on type I, class 3 extractors.

Nominal tap size on type II tap extractors.

3.5 Design and dimensions. Extractors shall be essentially the standard product of the manufacturer. A square end or other suitable means shall be provided for using a standard wrench in backing out broken screws and taps. Dimensions not having a minimum, maximum, or tolerance range, are considered to be nominal dimensions. Size deviations that are normal to standard manufacturing practices will be acceptable.

3.6 Sets. When sets of extractors are specified (see 6.2), the parts of the sets shall conform to the requirements of the respective items covered in this specification. When specified (see 6.2), extractor sets containing type III, screw extractors, sizes 1 through 5 shown in table VII, shall be furnished with drill guides, shown in table VIII. A drill shall be furnished for each extractor of the size recommended in table XV, column 2. The drill shall conform to GGG–D–751, type B, class 1, except for length. Length to be as furnished by the manufacturer. When specified (see 6.2), extractor sets containing type I, class 3, pipe extractors, sizes 1 through 6, shown in table IV, and type III, screw extractors, sizes 1 through 8, shown in table VII, shall be furnished with drill guides, shown in table VIII. A drill shall be furnished for each extractor of the size recommended in table XV, column 2. The drill shall conform to GGG–D–751, type B, class 1, except for length. Length to be as furnished by the manu-



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

facturer. A drill guide assembly in accordance with 3.14.3 shall also be furnished. Each set shall be furnished in a suitable container in accordance with 3.6.1, and as specified in the procurement document (see 6.2).

3.6.1 Wraps, roll, boxes, or blocks for sets. Wraps or rolls shall be of durable plastic or fabric with individual pockets for each extractor and its components, arranged to fold in a manner to enclose the extractor and its components, and shall be secured with durable snaps or ties. When boxes for sets are specified, they shall be either metal, plastic, or wood, with partitions for holding the components and a cover, having a positive or friction-type catch. When wooden blocks are specified, they shall have round drilled holes, to accommodate each extractor in the set. The interior and exterior surfaces of metal, wooden boxes, or wooden blocks shall have a suitable protective coating or varnish or other appropriate coating. When plastic materials are furnished in wraps, rolls, or boxes, the plastic shall be of a slow burning, selfextinguishing type (see 4.4.3).

3.7 Type I screw and pipe extractors.

3.7.1 Hardness. Type I extractors shall be heat-treated, and shall conform to the following Rockwell C hardness when tested as specified in 4.4.1:

| | Rockwell C |
|---|---|
| Type I, classes 1 and 2 | 52 to 60 |
| Type I, class 3 | 50 to 60 |

3.7.2 Torsional moment. Type I extractors shall conform to the minimum torsional moment specified in table I, II, III, IV, and V when tested as specified in 4.4.2.

3.8 Type I, class 1, styles A and B screw and pipe extractors, tapered, spiral flutes or ridges, drill and plug. The extractors shall have left-hand spiral flutes or ridges unless otherwise specified (see 6.2). The flutes shall start at small end, and generate working edges at least to the minimum length of specified taper.

3.9 Type I, class 1, style A, screw and pipe extractors, tapered, spiral flutes or ridges. Extractors shall conform to table I, and shall be similar to figure 1.



Figure 1. Type I, class 1, style A screw and pipe extractor, tapered, spiral flutes or ridges, drill. (for informational data, see 6.3).

Table I. Type I, class 1, style A, dimensional range and requirements.

| Extractor | Diameter small end | Diameter large end | Length of taper | Overall length | Torsional moment minimum |
|---|---|---|---|---|---|
| | Inches | Inches | Inches | Inches | Inch-pounds |
| 1 | 0.054 to 0.070 | 0.117 to 5/32 | 1/2 to 7/8 | 1-13/16 to 2-3/16 | 30 |
| 2 | 5/64 to 3/32 | .174 to 13/64 | 11/16 to 1-1/16 | 2-3/16 to 2-9/16 | 60 |
| 3 | 7/64 to 9/64 | 15/64 to 17/64 | 13/16 to 1-3/16 | 2-9/16 to 2-15/16 | 144 |
| 4 | 11/64 to 13/64 | 5/16 to 11/32 | 15/16 to 1-5/16 | 2-13/16 to 3-3/16 | 660 |
| 5 | 15/64 to 17/64 | 27/64 to 15/32 | 1-11/32 to 1-29/32 | 3-3/16 to 3-9/16 | 1,500 |
| 6 | 11/32 to 3/8 | 19/32 to 5/8 | 1-1/2 to 2 | 3-9/16 to 3-15/16 | 3,000 |
| 7 | 7/16 to 1/2 | 3/4 to 7/8 | 1-15/16 to 2-5/16 | 3-15/16 to 4-5/16 | 5,600 |
| 8 | 23/32 to 3/4 | 63/64 to 1-3/64 | 1-15/16 to 2-5/16 | 4-3/16 to 4-5/8 | 16,800 |
| 9 | 31/32 to 1 | 1-1/4 to 1-5/16 | 2-1/16 to 2-7/16 | 4-7/16 to 5-1/32 | 30,000 |
| 10 | 1-7/32 to 1-1/4 | 1-1/2 to 1-9/16 | 2-5/16 to 2-3/4 | 4-13/16 to 5-1/4 | 48,000 |
| 11 | 1-15/32 to 1-1/2 | 1-13/16 to 1-7/8 | 2-13/16 to 3-1/4 | 5-7/16 to 5-7/8 | 60,000 |
| 12 | 1-27/32 to 1-7/8 | 2-1/4 to 2-3/8 | 3-5/16 to 3-11/16 | 6-1/16 to 6-1/2 | 60,000 |



Source: https://assist.dla.mil -- Downloaded: 2026l0-26T12:43Z
Check the source to verify that this is the current version before use.

3.10 Type I, class 1, style B, screw and pipe extractors, tapered, spiral flutes or ridges, plug. Extractors shall conform to table II, and shall be similar to figure 2.

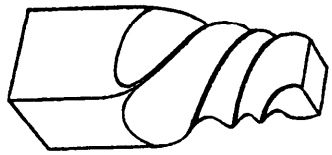


Figure 2. Type I, class 1, style B, screw and pipe extractor, tapered, spiral flutes or ridges, plug. (For informational data, see 6.3).

Table II. Type I, class 1, style B, dimensions and requirements.

| Extractor size number | Nominal dimension across corners, small end | Nominal dimensions across corners, large end | Minimum length of taper | Minimum overall length | Torsional moment, minimum |
|---|---|---|---|---|---|
| | Inches | Inches | Inches | Inches | Inch-pounds |
| 1 | 3/16 | 7/16 | 3/4 | 1- 1/2 | 650 |
| 2 | 1/4 | 17/32 | 13/16 | 1- 5/8 | 1,500 |
| 3 | 3/8 | 45/64 | 15/16 | 1- 3/4 | 2,400 |
| 4 | 13/32 | 7/8 | 1- 1/16 | 1-15/16 | 3,000 |
| 10 | 15/32 | 1- 1/16 | 1- 7/16 | 2- 5/8 | 4,000 |
| 5 | 11/16 | 1- 1/8 | 1- 9/16 | 2- 5/8 | 6,800 |
| 6 | 7/8 | 1- 9/64 | 1-13/16 | 2-13/16 | 14,000 |
| 7 | 1- 5/32 | 1-15/16 | 2- 3/16 | 3- 3/8 | 30,000 |
| 8 | 1-13/32 | 2- 1/8 | 2- 5/16 | 3- 1/2 | 60,000 |
| 9 | 2- 1/32 | 2-53/64 | 3- 1/4 | 4- 1/2 | 60,000 |

3.11 Type I, class 2, screw and pipe extractors, tapered, straight fluted, full length of taper. Extractors shall conform to table III, and shall be similar to figure 3.



Figure 3. Type I, class 2, screw and pipe extractor, tapered, straight fluted. (For informational data, see 6.3).

Table III. Type I, class 2, nominal dimensions and requirements.

| Extractor size number | Across corners dimension, small end | Across flats dimension, large end | Length of taper ±1/16 | Overall length | Torsional moment, minimum |
|---|---|---|---|---|---|
| | Inch | Inch | Inches | Inches | Inch-pounds |
| 1 | 1/8 | 3/16 | 1-13/32 | 2-5/16 | 90 |
| 2 | 11/64 | 3/16 | 1-29/32 | 2-5/8 | 160 |
| 3 | 13/64 | 1/4 | 1-11/32 | 2-7/8 | 320 |
| 4 | 19/64 | 5/16 | 1-7/16 | 3 | 500 |
| 5 | 11/32 | 3/8 | 1-5/8 | 3-1/4 | 670 |
| 6 | 27/64 | 7/16 | 1-23/32 | 3-7/8 | 900 |
| 7 | 1/2 | 1/2 | 2-13/32 | 4-1/2 | 1,250 |
| 8 | 19/32 | 5/8 | 2-9/16 | 4-7/8 | 1,650 |
| 9 | 23/32 | 3/4 | 2-15/32 | 5-3/8 | 2,100 |
| 10 | 31/32 | 1 | 3 | 5-1/2 | 3,500 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

3.12 Type I, class 3, pipe or screw extractors, straight, straight flutes, square tang. The extractors shall conform to table IV and V, and shall be similar to figure 4, and shall be provided with a lead taper for easy entry.

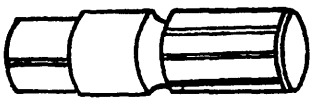


Figure 4. Type I, class 3, pipe or screw extractor, straight, straight fluted, square tang.

Table IV. Type I, class 3, for standard pipe or screw; dimensions and requirements.

| Extractor size number | Nominal standard pipe size | Diameter over flutes | Body diameter between flutes | Length of flutes ±1/16 | Overall length ±1/16 | Torsional moment, minimum | Size drill for torsional test |
|---|---|---|---|---|---|---|---|
| | Inch | Inches | Inches | Inches | Inches | Inch-pounds | Inches |
| 1 | 1/8 | 0.286 to 0.290 | 0.244 to 0.250 | 3/4 | 2-1/4 | 600 | 1/4 |
| 2 | 1/4 | .387 to .391 | .341 to .347 | 27/32 | 2-1/2 | 1,500 | 23/64 |
| 3 | 3/8 | .520 to .524 | .466 to .472 | 29/32 | 2-3/4 | 3,000 | 31/64 |
| 4 | 1/2 | .653 to .657 | .581 to .588 | 1 | 3 | 5,000 | 39/64 |
| 5 | 3/4 | .858 to .862 | .782 to .790 | 1 | 3-1/4 | 10,000 | 13/16 |
| 6 | 1 | 1.092 to 1.096 | 1.008 to 1.016 | 1 | 3-1/2 | 20,000 | 1- 1/32 |
| 7 | 1-1/4 | 1.433 to 1.437 | 1.332 to 1.340 | 1- 1/4 | 3-3/4 | 25,000 | 1-23/64 |
| 8 | 1-1/2 | 1.672 to 1.676 | 1.562 to 1.570 | 1- 3/8 | 4 | 30,000 | 1-19/32 |
| 9 | 2 | 2.141 to 2.146 | 2.014 to 2.022 | 1- 1/2 | 4-1/2 | 40,000 | 2- 3/64 |

Table V. Type I, class 3, for extra strong pipe; dimensions and requirements.

| Extractor size number | Nominal extra strong pipe size | Diameter over flutes | Body diameter between flutes | Length of flutes ±1/16 | Overall length ±1/16 | Torsional moment minimum | Size drill for torsional test |
|---|---|---|---|---|---|---|---|
| | Inch | Inches | Inches | Inches | Inches | Inch-pounds | Inches |
| 10 | 1/8 | 0.225 to 0.229 | 0.183 to 0.189 | 3/4 | 2-1/4 | 220 | 13/64 |
| 11 | 1/4 | .318 to .322 | .272 to .278 | 27/32 | 2-1/2 | 700 | 19/64 |
| 12 | 3/8 | .443 to .447 | .389 to .396 | 29/32 | 2-3/4 | 1,700 | 13/32 |
| 13 | 1/2 | .569 to .573 | .499 to .507 | 1 | 3 | 5,000 | 17/32 |
| 14 | 3/4 | .769 to .773 | .685 to .693 | 1 | 3-1/4 | 10,000 | 23/32 |
| 15 | 1 | .994 to .998 | .910 to .916 | 1 | 3-1/2 | 20,000 | 15/16 |
| 16 | 1-1/4 | 1.325 to 1.329 | 1.223 to 1.231 | 1-1/4 | 3-3/4 | 25,000 | 1-17/64 |
| 17 | 1-1/2 | 1.556 to 1.560 | 1.445 to 1.453 | 1-3/8 | 4 | 30,000 | 1-31/64 |
| 18 | 2 | 2.007 to 2.012 | 1.879 to 1.887 | 1-1/2 | 4-1/2 | 40,000 | 1-15/16 |

3.13 Type II, tap extractors. Type II tap extractors shall consist of a body, sleeve, collar, and fingers.

3.13.1 Body. The body shall be grooved in such a manner as to provide a sliding fit for the fingers. The body diameter shall be of such size that it can be inserted in the applicable tapped hole with a minimum amount of looseness.



Source: https://assist.dla.mil -- Downloaded: 202640-26T12:43Z
Check the source to verify that this is the currentversion before use.

3.13.2 Sleeve. The sleeve shall provide support for the fingers when removing broken taps, and shall be a sliding fit over the entire length of the assembled extractor.

3.13.3 Collar. The collar shall be a sliding fit over the entire length of the extractor and designed for the fingers to be easily detached or attached thereto.



Figure 5. Type II, tap extractor.

Table VI. Type II, tap extractor.

For standard machine and handtaps

| Extractor[1] size No. | Nominal tap size | Thread series | Number of flutes |
|---|---|---|---|
| | Inches | | |
| 4 | 0.112 | -- | 3 |
| 5 | .125 | -- | 3 |
| 6 | .138 | -- | 3 |
| 8 | .164 | -- | 4 |
| 10 | .190 | -- | 4 |
| 12 | .216 | -- | 4 |
| 1/4 | .250 | -- | 4 |
| 5/16 | .3125 | -- | 4 |
| 3/8 | .375 | -- | 4 |
| 7/16 | .4375 | -- | 4 |
| 1/2 | .500 | -- | 4 |
| 9/16 | .5625 | -- | 4 |
| 5/8 | .625 | -- | 4 |
| 11/16 | .6875 | -- | 4 |
| 3/4 | .750 | -- | 4 |
| 7/8 | .875 | -- | 4 |
| 1 | 1.000 | -- | 4 |
| 1-1/8 | 1.125 | NC | 4 |
| 1-1/8 | 1.125 | NF | 6 |
| 1-1/4 | 1.250 | NC | 4 |
| 1-1/4 | 1.250 | NF | 6 |
| 1-3/8 | 1.375 | NC | 4 |
| 1-3/8 | 1.375 | NF | 6 |
| 1-1/2 | 1.500 | NC | 4 |
| 1-1/2 | 1.500 | NF | 6 |

For standard pipe taps only

| Specify "pipe tap extractor" size | Specify number of flutes required | |
|---|---|---|
| 1/8 inch | 4 | 5 |
| 1/4 inch | 4 | 5 |
| 3/8 inch | 4 | 5 |
| 1/2 inch | 4 | 5 |
| 3/4 inch | | 5 |
| 1 inch | | 5 |

[1] Type II extractors not listed in table VI may be ordered by specifying the tap size, threads per inch, number of flutes, and kind of tap the extractor is for (for example: ½–13, three fluted, gun tap) (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

3.13.4 Fingers. The fingers shall be of high grade material and heat-treated so as to satisfactorily back out taps which are broken during normal production operations. The fingers shall be shaped to fit both the grooves in the body and the flutes of the applicable tap.

3.13.5 Hardness. Type II extractors, bodies and fingers shall be heat-treated, and shall conform to Rockwell C 50 to 58 hardness when tested as specified in 4.4.1.

3.13.6 Repair parts. All parts having the same manufacturer's part number shall be so constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.13.7 Type II tap extractors shall conform to table VI, and shall be similar to figure 5.

3.14 Type III, screw extractors. Type III screw extractors shall be straight, straight fluted. Sizes 1 through 5 shall consist essentially of a fluted rod with a sliding turnout nut, and shall conform to table VII, and shall be similar to figure 6. Sizes 6 through 8 shall be with square tang, and shall conform to table VII, and shall be similar to figure 4. All extractors shall have a lead taper for easy entry into the hole.



Figure 6. Type III, screw extractor, straight, straight fluted, sliding turnout nut.

Table VII. Type III, screw extractor, dimensions and requirements.

| Extractor size number | Diameter over flutes | Body Diameter between flutes | Nominal overall length ±1/16 | Torsional moment, minimum |
|---|---|---|---|---|
| | Inch | Inch | Inches | Inch-pounds |
| 1 | 0.132 to 0.139 | 0.119 to 0.124 | 2-5/16 | 55 |
| 2 | .198 to .205 | .181 to .186 | 2-3/4 | 210 |
| 3 | .266 to .272 | .244 to .249 | 3-1/16 | 560 |
| 4 | .335 to .341 | .296 to .302 | 3-5/16 | 1,120 |
| 5 | .371 to .377 | .326 to .332 | 3-3/4 | 1,370 |
| 6 | .461 to .467 | .413 to .419 | 4-1/8 | 1,780 |
| 7 | .525 to .531 | .471 to .478 | 4-1/2 | 3,500 |
| 8 | .651 to .657 | .585 to .592 | 4-3/4 | 7,500 |

3.14.1 Hardness. Type III extractors shall be heat-treated, and shall conform to Rockwell C 50 to 60 hardness when tested as specified in 4.4.1.

3.14.2 Drill guides for type III, sizes 1 through 5 screw extractors. Drill guides for use in centering and drilling lead holes shall be of carbon steel hardened and plated, (see 6.2). Drill guides shall be counterbored to allow for chip clearance. They shall conform to table VIII, and shall be similar to figure 7.

3.14.3 Drill Guide Assembly for use with type III screw extractors. The Drill Guide Assembly shall consist of a guide, nut, spacers, bushings, and a 1/4-in. diameter, 6-inch long, twist drill. The spacers and bushing shall be of carbon steel, hardened, and plated, and shall conform to table IX. The Drill Guide Assembly shall be similar to figure 8.

3.15 Workmanship. The workmanship shall be in accordance with high quality, commercial standards. Extractors shall be free from imperfections of any nature which may adversely affect the durability and serviceability.



Source: https://assist.dla.mil -- Downloaded: 202610-26T12:43Z
Check the source to verify that this is the current version before use.

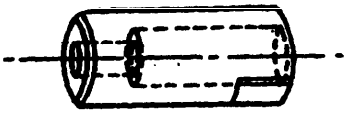


Figure 7. Drill guide for type III, sizes 1 through 5 screw extractors.

Table VIII. Drill guides for type III, sizes 1 through 5 screw extractors.

| Nominal outside diameter | Inside diameter | Nominal overall length |
|---|---|---|
| Inch | Inch | Inches |
| 9/32 | 0.125 to 0.131 | 1-3/16 |
| 5/16 | .125 to .131 | 1-3/16 |
| 11/32 | .125 to .131 | 1-3/16 |
| 3/8 | .125 to .131 | 1-3/16 |
| 13/32 | .1875 to .193 | 1-3/16 |
| 7/16 | .1875 to .193 | 1-3/16 |
| 15/32 | .1875 to .193 | 1-3/16 |
| 1/2 | .1875 to .193 | 1-3/16 |
| 17/32 | .250 to .256 | 1-3/16 |
| 9/16 | .250 to .256 | 1-3/16 |

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL–STD–105. Data for sampling shall be as stated in table X.

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3, as applicable.

4.4.1 Hardness test. The Rockwell hardness tests shall be conducted in accordance with method 243.1 of Fed. Std. No. 151.



Source: https://assist.dla.mil -- Downloaded: 202640-26T12:43Z
Check the source to verify that this is the current version before use.

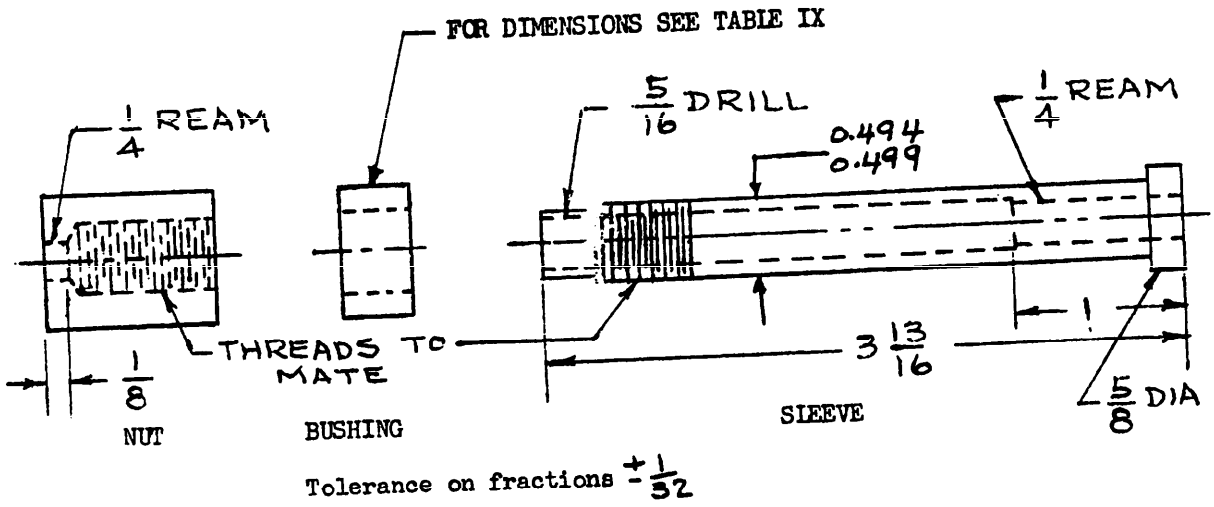



Figure 8. Assembly type drill guide for type III screw extractors.

Table IX. Sleeve bushings for assembly type drill guide, for type III screw extractors.

| Nominal outside diameter of bushing | Inside diameter | Nominal overall length | Quantity |
|---|---|---|---|
| Inch | Inch | Inch | |
| *5/8 | 0.501 to 0.506 | 3/4 | 4 |
| 21/32 | .501 to .506 | 3/8 | 2 |
| 11/16 | .501 to .506 | 3/8 | 2 |
| 23/32 | .501 to .506 | 3/8 | 2 |
| 25/32 | .501 to .506 | 3/8 | 2 |
| 13/16 | .501 to .506 | 3/8 | 2 |
| 27/32 | .501 to .506 | 3/8 | 2 |
| 29/32 | .501 to .506 | 3/8 | 2 |
| 15/16 | .501 to .506 | 3/8 | 2 |
| 31/32 | .501 to .506 | 3/8 | 2 |
| 1-1/32 | .501 to .506 | 3/8 | 2 |
| 1-1/16 | .501 to .506 | 3/8 | 2 |
| 1-3/32 | .501 to .506 | 3/8 | 2 |

* Spacers

4.4.2 Torsional moment test. Each of the sample extractors shall be tested in the following manner:

4.4.2.1 A hole shall be drilled in a steel plate or block, the hole size being determined by referring to the largest drill size for the specific type, class, style, and size of extractor listed in the applicable table of this specification. The steel plate or block shall be of carbon tool steel heat-treated to a Rockwell hardness of 36 to 42 on the C scale when testing the following extractors:

Type I, class 1, style A, sizes 7 through 12, inclusive;
Type I, class 1, style B, sizes 6 through 9, inclusive.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

Table X. Sampling data.

| Category | Sample unit | Inspection level | Acceptable quality level | AQL expressed in terms of | Reference |
|---|---|---|---|---|---|
| Extractor with or without components: Visual examination | 1 ea. | II | 4.0 | Defects per hundred units | 4.3.1 |
| Dimensional examination | 1 ea. | S-4 | 1.5 | Defects per hundred units | 4.3.2 |
| Testing | 1 ea. | S-2 | 1.0 | Defects per hundred units | 4.4.1 and 4.4.2 |
| Sets: Visual examination | 1 ea. | S-3 | 1.5 | Defects per hundred units | See note |
| Testing | 1 ea. | S-1 | 1.0 | Percent defective | See note and 4.4.3 |
| Preparation for delivery | One container | S-2 | 4.0 | Defects per hundred units | 4.5 |

NOTE: Components of sets shall have been inspected and accepted prior to assembly into set. Sets shall be checked for completeness, damage of components, and compliance with 3.6 and 3.6.1. Plastic wraps, rolls, or boxes shall be tested for flammability (see 4.4.3).

The steel plate or block shall be of medium carbon steel heat-treated to a Rockwell hardness reading of 24 to 30 on the C scale when testing the following extractors:

Type I, class 1, style A, sizes 4 through 6, inclusive;
Type I, class 1, style B, sizes 2 through 5, inclusive, and including size 10;
Type I, class 2, sizes 4 through 10, inclusive.

When testing extractors not included in the above group, the steel plate or block shall be of low-carbon unhardened steel. If lead holes are not drilled, care shall be exercised to select drills that are ground on center, in order to prevent the holes from being drilled oversize. A new hole is required for each extractor when testing type I, classes 1 and 2. The extractor shall be inserted in the drilled hole and gently driven in sufficiently to provide a firm engagement. A torque testing machine or other suitable means shall be utilized to exert a steady pressure of torque until the minimum torsional moment in inch-pounds is reached or the extractor fails. Failure shall be deemed to have occurred when the extractor breaks, or a permanent bend or twist occurs. If more than one of the sample extractors fail to meet the minimum inch-pounds specified under torsional moment specified in table I, II, III, IV, V, and VII, as applicable, this shall be cause for rejection of the lot represented by the sample.

4.4.3 Flammability test. Plastic items shall pass the flammability test. This test shall be in accordance with method 2022 of Fed. Test Method Std. No 406.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table XI. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2) the extractors shall be preserved, packaged and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C, and the level of packing A, B, or C, as specified (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 202640-26T12:43Z
Check the source to verify that this is the current version before use.

5.1.1 As part of the requirements of section 5 of this specification, table XIA is considered part of and used in lieu of that portion of table I of MIL–H–15424 covering Extractor, Screw (Stud and Pipe), and Extractor, Tap.

Table XI. Classification of preparation for delivery defects.

| Examine | Defects |
|---|---|
| Markings (exterior and interior) | Omitted; incorrect; illegible; improper size, location, sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container. |

Table XIA. Packaging of hand tools.

| Column 1 | Column 2 | Column 3 | Column 4 | Column 5 | Column 6 | Column 7 |
|---|---|---|---|---|---|---|
| Extractor: | | | | | | |
| Screw and pipe. | | IC–1 | | | | |
| Up through 1/2-inch drill size. | P–7,P–10 | IC–1 | 6 | 8 | 6 | 288 |
| 5/8-inch drill size through 2-1/8-inch drill size. | P–7,P–10 | IC–1 | 2 | | 2 | 4 |
| Tap: | | | | | | |
| Up through 1/4-inch. | P–7,P–10 | IC–1 | 6 | | 16 | 96 |
| 5/16 inch through 7/8 inch. | P–7,P–10 | IC–1 | 6 | | 6 | 36 |
| 1 inch through 1-1/2 inch. | P–7,P–10 | IC–1 | 3 | | 6 | 18 |
| Screw: | | | | | | |
| Up through 3/4 inch. | P–7,P–10 | IC–1 | 6 | | 60 | 360 |

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I extractors are not intended to be used in removing broken studs or tight-fitting threaded parts that have been subjected to heat and corrosion over a long period of time unless the threads were covered with a nonseizing compound such as a graphite-and-oil mixture.

6.1.2 Type I, class 1, styles A and B, and type 1, class 2. Tables XII, XIII, and XIV list drill sizes which should be used for the applicable extractors in removing the listed bolt and screw sizes under normal conditions. However, under unusual conditions, the use of other size drills for broken studs and screws is sometimes required, depending on the length of the broken section and its depth in the hole. In general, the largest possible extractor should be used, and when the root diameter of the threaded part will permit, it is advisable to use larger drill sizes so the stress will be upon the larger diameter of extractor.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

6.1.3 Type I, class 3. Type I, class 3, extractor is used primarily for removing broken pipe nipples and fittings.

6.1.4 Type II. Type II tap extractors are not expected to remove all broken taps, especially if the tap is dull and has broken after reaching considerable depth in a material that lacks nonseizing properties. Most taps that are broken during normal production operations may be removed, although assistance from a penetrating compound and jarring of the broken section with a blunt punch and hammer is sometimes required.

6.1.5 Type III, sizes 1 through 5. Type III, sizes 1 through 5 extractors may be used for removing broken head and manifold bolts or studs on gasoline-engine blocks without removing the head or manifold. When used in this manner, drill a centering hole through the broken end using a close-fitting drill guide, remove the guide and drill the correct size hole to receive the extractor, remove the chips and after lubricating the extractor, drive it in the hole and slide the turnout nut close to the head or manifold, then with a suitable wrench back the broken section out.

6.2 Ordering data. Purchaser should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style (if applicable) of extractor required (see 1.2).
(c) Whether sets are to be furnished in plastic or fabric wraps or rolls, or metal, plastic, or wooden boxes, or wooden blocks (see 3.6 and 3.6.1).
(d) If type I, class 1, style A or B extractors are required for removing broken screws having left-hand threads, specify right-hand spiral flutes or ridges (see 3.8).
(e) Size of extractors or drill guides required (see tables I, II, III, IV, V, VI, VII, VIII, and IX).
(f) When type II extractors, for machine and handtaps are ordered, specify the number of flutes required on size 1-1/8-inch and above (see table VI).
(g) When type II extractors, for pipe taps are ordered, specify "pipe tap extractor" and number of flutes required as applicable, (see table VI).
(h) When type III screw extractors, are ordered in sets, specify whether sizes 1 through 5, or sizes 1 through 8 are required. Specify if with drill guides (3.14.2) and drills (table XV), also if assembly type drill guide with bushings (3.14.3) is required.
(i) Selection of applicable level of preservation, packaging, and packing required (see 5.1 and 5.2).
(j) Any special marking for shipment required (see 5.2).

Table XII. Type I, class 1, style A.

| Extractor size number | Use drill size | For screw and bolt screw | Standard pipe size |
|---|---|---|---|
| | Inches | Inches | Inches |
| 1 | 1/16 to 5/64 | 3/16 to 1/4 | ------ |
| 2 | 3/32 to 7/64 | 1/4 to 5/16 | ------ |
| 3 | 9/64 to 5/32 | 5/16 to 7/16 | ------ |
| 4 | 7/32 to 1/4 | 7/16 to 9/16 | ------ |
| 5 | 17/64 to 19/64 | 9/16 to 3/4 | 1/8 to 1/4 |
| 6 | 25/64 to 27/64 | 3/4 to 1 | 3/8 |
| 7 | 33/64 to 35/64 | 1 to 1-3/8 | 1/2 |
| 8 | 25/32 to 27/32 | 1-3/8 to 1-3/4 | 3/4 |
| 9 | 1-1/16 to 1-1/8 | 1-3/4 to 2-1/8 | 1 |
| 10 | 1-9/32 to 1-11/32 | 2-1/8 to 2-1/2 | 1-1/4 |
| 11 | 1-17/32 to 1-19/32 | 2-1/2 to 3 | 1-1/2 |
| 12 | 1-15/16 to 2 | 3 to 3-1/2 | 2 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

Table XIII. Type I, class 1, style B.

| Extractor size number | Use drill size | For screw and bolt screw | Standard pipe size |
|---|---|---|---|
| | Inches | Inches | Inches |
| 1 | 13/64 to 15/64 | 3/8 | 1/8 |
| 2 | 9/32 to 19/64 | 1/2 | 1/4 |
| 3 | 25/64 to 27/64 | 5/8 to 3/4 | 3/8 |
| 4 | ---------- | -------- | 1/2 |
| 10 | 1/2 to 17/32 | 7/8 to 1 | -- |
| 5 | 23/32 to 3/4 | 1-1/8 to 1-1/4 | 3/4 |
| 6 | 29/32 to 59/64 | 1-1/2 to 1-3/4 | 1 |
| 7 | 1-3/16 | 2 | 1-1/4 |
| 8 | 1-7/16 to 1-15/32 | 2-1/4 to 2-1/2 | 1-1/2 |
| 9 | 2-1/16 | 3 to 3-1/2 | 2 |

Table XIV. Type I, class 2.

| Extractor size number | Use drill size | Recommended depth of drill hole | For screw and bolt size | Standard pipe size |
|---|---|---|---|---|
| | Inches | Inch | Inches | Inch |
| 1 | 1/8 | 3/8 | 1/4 to 5/16 | -- |
| 2 | 3/16 | 1/2 | 3/8 | -- |
| 3 | 15/64 | 1/2 | 7/16 | 1/8 |
| 4 | 5/16 | 1/2 | 1/2 to 9/16 | 1/4 |
| 5 | 3/8 | 5/8 | 5/8 | 3/8 |
| 6 | 15/32 | 11/16 | 3/4 | -- |
| 7 | 9/16 | 11/16 | 7/8 | 1/2 |
| 8 | 21/32 | 3/4 | 1 to 1-1/8 | 1/2 |
| 9 | 13/16 | 3/4 | 1-1/4 to 1-3/8 | 3/4 |
| 10 | 1-1/16 | 3/4 | 1-1/2 | 1 |

Table XV. Type III.

| Extractor size number | Use drill size | For screw size |
|---|---|---|
| | Inch | Inch |
| 1 | 1/8 | 1/4 |
| 2 | 3/16 | 5/16 |
| 3 | 1/4 | 3/8 |
| 4 | 5/16 | 7/16 |
| 5 | 11/32 | 1/2 |
| 6 | 7/16 | 5/8 |
| 7 | 1/2 | 3/4 |
| 8 | 5/8 | 1 |

6.2.1 When ordering repair parts for type II tap extractors, specify the following:

(a) The manufacturer's name or trademark of extractor for which the parts are required.

(b) The name of the component part.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

(c) The name of the component part.
(d) Quantity of each item to be procured.
(e) The tap size, threads per inch, number of flutes, and kind of tap the extractor is for.

6.3 Informational data. The informational data contained in table XII through XV is applicable to tables I through III and VII respectively.

6.4 Civil agency procurement. When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

6.5 Supression data. The classification of extractors covered in this specification corresponding to those covered in GGG–E–936a are as follows:

| GGG–E–936a | GG–E–00936c |
|---|---|
| Type I. Screw and pipe extractors.<br>Class 1. Tapered, spiral flutes, or ridges.<br>Style A. Drill.<br>Style B. Plug.<br>Class 2. Tapered, straight fluted.<br>Class 3. Straight, straight fluted.<br>Style A. Sliding turn out nut.<br>Style B. Square tang.<br><br>Type II. Tap extractors. | Type I. Screw and pipe extractors.<br>Class 1. Tapered, spiral flutes, or ridges.<br>Style A. Drill.<br>Style B. Plug.<br>Class 2. Tapered, straight fluted.<br>Class 3. Straight, straight fluted.<br>Type III. Screw extractors, sizes 1 through 5.<br>Type III. Screw extractors. Sizes 6 through 8 (from table IV GGG–E–936a only).<br>Type II. Tap extractors. |

DOD coordination has been waived.

Preparing Activity:

GSA–FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of the specification to obtain extra copies and other documents referenced herein. Price 10 cents each.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.